Ahmed Marnissi

Embedded Software Engineer

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Summary

Embedded Software Engineer with 3 years of experience developing real-time and Linux-based embedded systems. Specializing in safety-critical applications and low-latency protocol design, with strong expertise in C/C++, QNX/FreeRTOS, and communication stacks.

Technical Skills

Programming: C, C++, Python
Operating Systems: Embedded Linux, QNX RTOS, FreeRTOS
Architectures: ARM Cortex-M4/M7, Cortex-A7
Platforms: STM32, STM32MP1, i.MX
Networking & Protocols: TCP/IP, UDP, Ethernet, CAN, CANopen, I^2C, SPI, UART, I2S
Middleware & IPC: LwIP, OpenAMP, POSIX IPC, Multi-threading
DSP Techniques: FIR Filtering, FFT, Decimation
Tools: Git, SVN, Jenkins, Docker, Jira, Confluence
Testing & Debugging: LDRA, GoogleTest, J-Link, Logic Analyzer, HackRF

Professional Experience

Embedded Software Engineer | ***C/C++, Linux, QNX, FreeRTOS*** — Aug 2023 – Present
ACTIA Engineering Services — *Tunis, Tunisia*

- **Crane Anti-Collision System:**
- Engineered TDM over UDP protocol in C/C++ achieving < 50 ms latency over 800 m using Wi-Fi HaLow.
- Implemented LTE TCP fallback communication ensuring reliable safety data exchange (~ 200 ms latency).
- Developed GNSS HTTP middleware enabling RTK Fix accuracy of ± 10 cm.
- Designed real-time processing pipeline on QNX RTOS achieving < 5 ms latency.
- Developed custom device drivers (ICM-42605 over I^2C) reducing hardware cost by 80%.
- Ported 32-bit Linux codebase to 64-bit architecture and Dockerized build environment.
- Applied V-cycle development methodology with unit testing and static analysis (LDRA).
- **Diagnostic Event Manager (DEM):**
- Developed a C++ Diagnostic Event Manager module aligned with AUTOSAR diagnostic specifications.
- Implemented event state handling, Diagnostic Trouble Code (DTC) management, and fault memory mechanisms.
- Developed a non-volatile storage module ensuring persistent DTC and event retention across resets and power cycles.

Embedded Software Intern | ***C/C++, Linux, Buildroot, OP-TEE, STM32MP*** — Feb 2023 – Jun 2023
ACTIA Engineering Services — *GitHub Project* — *Tunis, Tunisia*

- Built custom Embedded Linux image using Buildroot (kernel, U-boot, rootfs, cross-toolchain).
- Integrated ARM TrustZone and OP-TEE on STM32MP1 enabling secure/non-secure world execution.
- Implemented mailbox-based inter-core communication between Cortex-A7 (Linux) and Cortex-M4 (FreeRTOS).

Projects

Audio Decimation and Spectrum Analysis | *C, STM32, DSP, I2S, FFT*

- Designed DSP pipeline converting PDM data to PCM using multi-stage FIR-based decimation.
- Performed FFT analysis and built visualization tools for real-time frequency spectrum inspection.

RTSP Audio Streaming Client on STM32 | *C, STM32, FreeRTOS, LwIP*

- Developed RTSP/RTP client enabling real-time audio streaming over Ethernet.
- Designed deterministic multi-task architecture using FreeRTOS for reliable packet handling.

DTMF Audio Decoder on STM32 | *STM32, C, DSP*

- Implemented DTMF decoder using FIR band-pass filter bank targeting standard DTMF frequency pairs.
- Optimized fixed-point convolution achieving real-time tone detection with minimal CPU overhead.

Education

National Institute of Applied Sciences and Technologies (INSAT) — Tunis, Tunisia
Engineering Degree in Industrial Computing and Automation — *2018 – 2023*

Organizations

IEEE INSAT RAS Student Branch — Sep 2020 – Present
Member & Embedded Systems Trainer — *Tunis, Tunisia*

- Participated in the Eurobot International Robotics Competition contributing to embedded system design and integration.
- Conducted embedded systems workshops on microcontroller programming and communication protocols.

Technical Writing

Contributor — Embedded Systems Articles
Wadix Technologies

- Authored technical articles on ARM architecture, FreeRTOS, embedded Linux, and embedded software design.
- Blog: medium.com/@wadixtech

Awards

1st Place – TICAD8 National & African Automotive Hackathon
ACTIA Engineering Services and StartupGateX

- AI dashboard assistant with drowsiness detection using TinyML and OpenCV.
- *Skills:* TinyML, Computer Vision, Embedded AI

2nd Place – Industrie 4.0 Hackathon
Orange Digital Center

- End-to-end defect detection system with Raspberry Pi & Edge Impulse.
- *Skills:* IoT, AI, Edge Computing

2nd Place – BOT Camp Hackathon
Evident Industrial and IEEE INSAT RAS

- Mobile probe mount for inspection drones using ROS & SolidWorks.
- *Skills:* Robotics, ROS, NDT

Languages

English | **French** | **Arabic**